

Software Engineering Project Proposal

Course Code: CSE-3104 **Submission Date:** 15/5/26

Submitted to: Md Samsuddoha, Assistant Professor, Department of Computer Science & Engineering, University of Barishal

Team Formation

- **Team Name:** Team StormTroopers
- **Team Members and Roles:**
 - **Mohammad Abdullah Azrof Sadim (230102026):** Team Leader & Systems Architect - Project coordination, agile management, system architecture design, Git collaboration, and ensuring decoupled frontend/backend communication.
 - **Araf M N Ishtiaq (230102027):** Backend Engineer - Python (FastAPI) backend development, API routing, simulation loop creation, and managing the core transit logic.
 - **Yeasin Arafat (230102022):** AI & Machine Learning Engineer - Scikit-Learn model training, mathematical logic for "Self-Sharpening" traffic ETAs, and determining 100/200% capacity thresholds.
 - **Sajal Tikadar (230102031):** Frontend & Spatial UI Developer - Interface design, React.js implementation, Mapbox GL JS integration, and rendering interactive 3D bus models on the map.
 - **Partha Sarathi Biswas (230102013):** Database & Real-Time Systems - PostgreSQL database schema design, historical travel data logging, and managing Socket.IO WebSockets for zero-latency telemetry streaming.
 - **Sojib Ahmed (230102008):** QA Tester & Deployment Engineer - Writing automated Geofencing logic, bug tracking, system stress-testing, and deploying the application to cloud platforms.

SWOT Analysis of Team

- **Strengths:** Deep team understanding of decoupled architectures, WebSockets, and AI integration. The team directly understands the real cultural challenges (like 200% standing capacity) in local transit. Highly innovative fusion of 3D spatial computing and machine learning.
- **Weaknesses:** Lack of physical IoT hardware and enterprise funding for a full physical prototype. Strict 1-month development time due to academic coursework constraints.
- **Opportunities:** Fills a documented academic gap in Human-Computer Interaction (HCI) for Digital Twins. High educational impact by saving thousands of collective student commute hours. Potential to scale the MVP architecture to national commercial transit fleets.
- **Threats:** Unpredictable third-party API limits (Mapbox free tier). Potential network latency affecting real-time WebSocket data streaming.

Project Title

CampusFlow 3D - A Web-Based University Transit Digital Twin Platform

Introduction/Overview

The domain of this project is Smart Mobility and Intelligent Transportation Systems (ITS). CampusFlow 3D is a software-based Digital Twin engineered to digitize and centralize the logistics of the University of Barishal's transit network. The intended users are the university students and the transport pool administration. Through this platform, students can visually track the university's 20-bus fleet across six major routes via a live, 2.5D interactive WebGL spatial environment. The importance of this project lies in its shift from traditional, reactive transit management to a modern, predictive, and data-driven ecosystem. It provides administrators with a centralized dashboard for oversight, while giving students historically sharpened ETAs, dynamic route availability, and AI-driven capacity tracking to eliminate guesswork.

Problem Statement

In universities across Bangladesh, transit management processes remain completely static and largely blind. At the University of Barishal, buses operate via designated stops without live tracking, forcing students to guess arrival times based entirely on static, paper-based departure schedules. This disconnect creates a significant limitation known as the "Blind Waiting" phenomenon. Students regularly wait 30 to 40 minutes at major traffic bottlenecks without knowing the capacity of the approaching vehicle. Discovering an arriving bus is at 100% capacity with no standing room left results in severely wasted time, frequently causing students to miss urgent classes and exams. Additionally, the system suffers from "Ghost Bus" uncertainty; if a peripheral route is inactive or breaks down, students wait in vain. This problem is highly significant because the administration's lack of empirical data on when and where buses become overloaded makes efficient fleet optimization mathematically impossible.

Motivation

The motivation behind CampusFlow 3D comes from the daily, observable struggles of Barishal University students losing valuable time to unpredictable transit. This project will directly benefit students by empowering them to make executive, data-driven decisions. If a student sees an approaching bus glowing "Red" (overloaded), they can instantly arrange local transport instead of waiting blindly. Furthermore, this project acts as an academic incubator. While the initial focus is the university campus, the ultimate long-term vision is to scale this AI-driven Digital Twin to commercial poribohon bus services across Bangladesh. This brings potential innovations to inter-city travel by eliminating blind waits at ticket counters and establishing a unified, predictive smart-transit grid.

Expected Features/Requirements

- Browse a live, high-contrast 2.5D interactive Mapbox interface showing real-time spatial tracking of the active fleet.

- Click 3D bus models (.gltf) to view Route Designation, Number, and Next Stop details.
- View "Self-Sharpening" machine learning ETA predictions adjusted for rush-hour traffic.
- Identify overloaded buses instantly via AI-driven Capacity Color-Coding (Yellow for 100% seating, Red for 200% cultural standing room limit).
- Utilize automated Geofencing to toggle buses "In Service" or "Out of Service" automatically, keeping a persistent list of route availability.
- Access a centralized Transport Pool Dashboard for administrators to monitor the entire active 20-bus fleet simultaneously.
- Receive automated high-priority SOS alerts with locked GPS coordinates if a bus breaks down.
- View historical Analytics Dashboards to monitor average crowd density and generate fleet impact reports.

Development Approach

- **Model:** Agile (supporting iterative prototyping across distinct modules: Map rendering, Backend simulation, WebSockets, AI integration, and Analytics).
- **Tools & Technology:**
 - **Frontend:** React.js, Mapbox GL JS, Tailwind CSS (for high-performance component UI and 2.5D rendering).
 - **Backend:** Python (FastAPI) (for API routing and simulation loop processing).
 - **Database:** PostgreSQL (for logging historical travel data).
 - **AI / ML:** Scikit-Learn (for time-series ETA predictions and capacity thresholds).
 - **Real-Time:** Socket.IO / WebSockets (for zero-latency event streaming).

Budget Estimation

(Note: This project relies on a dual budget model, contrasting a theoretical enterprise reality against the actual Minimum Viable Product delivery.)

Theoretical Enterprise Rollout (20 Buses): 15,00,000 BDT

- **Hardware (6.4 Lakh BDT):** 4G Trackers, Passenger Cameras, and Edge Nodes.
- **Software & Infrastructure (4.0 Lakh BDT):** Commercial Licenses and AWS Hosting.
- **Labor & Maintenance (4.6 Lakh BDT):** Engineering, Technicians, and Spares.

Academic MVP (Engineering a 0 BDT Simulation Ecosystem)

- **Replacing Hardware (0 BDT):** Achieved through unpaid academic development and simulated Python telemetry loops.
- **Replacing Mapping Licenses (0 BDT):** Utilizing the Mapbox Developer Free Tier and Community 3D assets.
- **Replacing AWS Cloud (0 BDT):** Using Vercel for Frontend Edge Hosting and Render/PythonAnywhere for Backend/WebSocket Hosting.

Takeaway: We are leveraging modern free-tier cloud infrastructure to build and host an enterprise-grade Digital Twin architecture without any initial capital expenditure.

Timeline

- **Days 1-10 (Sprint 1 - Planning):** Project requirement analysis, Git initialization, React setup, and tracing GeoJSON coordinates for routes.
- **Days 11-25 (Sprint 2 - Simulation):** Python FastAPI backend development, database schema design, and creating simulation loops for moving coordinates.
- **Days 26-40 (Sprint 3 - Real-Time):** Mapbox integration, 3D model rendering, and establishing Socket.IO WebSockets to stream data to the UI.
- **Days 41-55 (Sprint 4 - Intelligence):** Scikit-Learn integration for Self-Sharpening ETA, Geofencing state logic, and Red/Yellow capacity triggers.
- **Days 56-65 (Testing):** Admin Dashboard creation, automated SOS visual UI integration, rigorous system debugging, and WebSocket latency testing.
- **Days 66-70 (Deployment):** Final cloud deployment, complete project documentation, presentation preparation, and final academic submission.

Conclusion

CampusFlow 3D is designed to address a critical operational challenge faced by the university campus: the reliance on static, blind transit dispatching. It brings a modern, proactive Digital Twin experience into the university environment. For students, it provides a simple way to verify route availability, track intelligent ETAs, and avoid overloaded buses. For administrators, it offers a professional dashboard to monitor the fleet and gather historical data for route optimization. In essence, CampusFlow 3D represents a highly feasible, deeply innovative vision: making transit easier to track, easier to manage, and more predictable for every student and administrator.